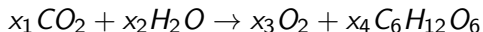


Photosynthesis



Carbon: $x_1 = 6x_4$

Oxygen: $2x_1 + x_2 = 2x_3 + 6x_4$

Hydrogen: $2x_2 = 12x_4$

$$x_1 + 0x_2 + 0x_3 - 6x_4 = 0$$

$$2x_1 + x_2 - 2x_3 - 6x_4 = 0$$

$$0x_1 + 2x_2 + 0x_3 - 12x_4 = 0$$

Opgavesamling FF503 Almen Kemi

D18

- 1) Der skal fremstilles 100 mL af en fosfatpuffer bestående af HPO_4^{2-} og H_2PO_4^- med $\text{pH} = 7,21$. Totalkoncentrationen af pufferen skal være 0,100 M (dvs. $[\text{HPO}_4^{2-}] + [\text{H}_2\text{PO}_4^-] = 0,100 \text{ M}$). Det kan anvendes, at $\text{pK}_a(\text{H}_2\text{PO}_4^-) = 7,21$.

Beregn hvor meget $\text{Na}_2\text{HPO}_4 \cdot 2\text{H}_2\text{O}$ og $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, der skal afvejes til dette formål.

Pufferligning:

$$\text{pH} = \text{pK}_a + \lg \frac{n_b}{n_s}$$

where n_b - amount of HPO_4^{2-} (hydrogen phosphate ion), n_s - amount of H_2PO_4^- (dihydrogen phosphate ion), pK_a - negative logarithm of the acid dissociation constant

More generally, we consider the pair of equations

$$a_{11}x + a_{12}y = b_1, \quad (1)$$

$$a_{21}x + a_{22}y = b_2 \quad (2)$$

(linear in the unknowns x and y).

$$(1) \cdot a_{22} - (2)a_{12} \implies$$

$$(a_{11}a_{22} - a_{12}a_{21})x = b_1a_{22} - a_{12}b_2.$$

Analogously:

$$(a_{11}a_{22} - a_{12}a_{21})y = a_{11}b_2 - b_1a_{21}.$$

With

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} := a_{11}a_{22} - a_{12}a_{21}, \quad D_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}, \quad D_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$$

we can rewrite these equations as

$$Dx = D_1, \quad Dy = D_2.$$

Example 1.1.

$$2x - 3y = 5$$

$$x + 5y = 9.$$

Example 1.2 (Poll everywhere).

What is the value of $\begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix}$?

As seen above, a **unique solution** exists $\iff D \neq 0$. The system is then called **well determined**. $D = 0$ implies $D_1 = 0$, $D_2 = 0$. In this case we have **infinitely many solutions** and the system is called **underdetermined**. If $D = 0$ but $D_1 \neq 0$ or $D_2 \neq 0$, we have **no solution** at all and the system is called **overdetermined**.

Example 1.3.

$$\begin{aligned}x + y &= 1 \\ -x - y &= 1.\end{aligned}$$

Example 1.4 (Poll everywhere).

Is

$$2x_1 + 5x_2 = 5, \quad -3x_1 + x_2 = 2$$

overdetermined, underdetermined, or well determined?

For a **homogeneous** system of equations it follows that there exists a nontrivial solution only if the determinant of the coefficients is zero.

Example .

$$D = \begin{vmatrix} 2 & 1 & 3 \\ 4 & -2 & 0 \\ -1 & 1 & 0. \end{vmatrix}$$

Biological objects of interest have structure that suggest to us the way those objects should be analyzed, for example:

- ▶ various components of cells, e.g. chemical composition
- ▶ composition of different individuals which make up a population, e.g. demographic structure (gender, age, location, economic status) or epidemiological structure (susceptible, infected, recovered)
- ▶ species which make up a community

In the following we will discuss mathematical concepts that provide a way to describe the structure/composition of these biological objects and a mechanism to analyze how the composition might change through time or across space (dynamics)

As motivating example we will look at a simple population model for Herpes simplex virus (HSV, the virus which causes herpes). We divide the population into three categories:

- (1) susceptible (does not have the virus),
- (2) infected and shedding the virus (can infect others),
- (3) quiescent (infected but not shedding the virus).

We must decide how to order them: We will use the order: susceptible, infected, quiescent.

If we let

- ▶ s = proportion of population that is susceptible,
- ▶ i = proportion of population that is infected and infective,
- ▶ q = proportion of population that is quiescent

then the vector v representing the epidemiological structure of the population would be:

$$v = \begin{pmatrix} s \\ i \\ q \end{pmatrix}.$$

E.g. $v = \begin{pmatrix} 0.65 \\ 0.20 \\ 0.15 \end{pmatrix}$ would mean that 65 % of the population are susceptible, 20% infected and infectious, and 15 % quiescent. Addition of vectors and multiplication with a scalar c (a real number): If $v, w \in \mathbb{R}^n$ (n-tuples of real numbers), then

$$v + w = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \\ \vdots \\ v_n + w_n \end{pmatrix}, \quad cv = \begin{pmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{pmatrix}.$$

Example 2.1.

Assuming that the population of Danmark is $c=6$ Mio. people, the number of people in the different HSV classes is

$$cv = 6 \cdot 10^6 \begin{pmatrix} 0.65 \\ 0.20 \\ 0.15 \end{pmatrix} = \begin{pmatrix} 3.9 \cdot 10^6 \\ 1.2 \cdot 10^6 \\ 0.9 \cdot 10^6 \end{pmatrix}.$$

Dynamics: Vectors changing over time

Assume that each day,

- ▶ 0.0054% of the susceptible population becomes infected and shedding the virus,
- ▶ 28.6% of those infected move into the quiescent stage,
- ▶ 2.9% of those in the quiescent stage start to shed the virus again.

How is the situation after one day, after a month, and after a year?

$$\begin{aligned}s(n+1) &= s(n) - 0.000054 \cdot s(n) \\ &= 0.999946 \cdot s(n),\end{aligned}$$

$$\begin{aligned}i(n+1) &= i(n) + 0.000054 \cdot s(n) - 0.286 \cdot i(n) + 0.029 \cdot q(n) \\ &= 0.000054 \cdot s(n) + 0.714 \cdot i(n) + 0.029 \cdot q(n),\end{aligned}$$

$$\begin{aligned}q(n+1) &= q(n) + 0.286 \cdot i(n) - 0.029 \cdot q(n) \\ &= 0.286 \cdot i(n) + 0.971 \cdot q(n).\end{aligned}$$

Rewriting, we have

$$s(n+1) = 0.999946 \cdot s(n) + 0.000 \cdot i(n) + 0.000 \cdot q(n),$$

$$i(n+1) = 0.000054 \cdot s(n) + 0.714 \cdot i(n) + 0.029 \cdot q(n),$$

$$q(n+1) = 0.000000 \cdot s(n) + 0.286 \cdot i(n) + 0.971 \cdot q(n).$$

If we form an array of only the numbers, we obtain the coefficient matrix

$$\begin{pmatrix} 0.999946 & 0 & 0 \\ 0.000054 & 0.714 & 0.029 \\ 0 & 0.286 & 0.971 \end{pmatrix}$$

- ▶ This is a (3×3) matrix).
- ▶ Each column sums to 1.
- ▶ Diagonal entries represent proportions of each class that stay in that class each time step
- ▶ Off-diagonal entries represent proportions of each class that move to other classes each time step

- The coefficient matrix is also called a transfer matrix, as it can be interpreted in the following way:

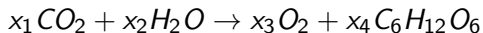
$$\begin{array}{rcc}
 & & \text{moving from} \\
 & & s(n) \quad i(n) \quad q(n) \\
 \text{moving into} \begin{array}{l} s(n+1) \\ i(n+1) \\ q(n+1) \end{array} & \begin{pmatrix} 0.999946 & 0 & 0 \\ 0.000054 & 0.714 & 0.029 \\ 0 & 0.286 & 0.971 \end{pmatrix}
 \end{array}$$

- We want to write:

$$\begin{pmatrix} s(n+1) \\ i(n+1) \\ q(n+1) \end{pmatrix} = \begin{pmatrix} 0.999946 & 0 & 0 \\ 0.000054 & 0.714 & 0.029 \\ 0 & 0.286 & 0.971 \end{pmatrix} \begin{pmatrix} s(n) \\ i(n) \\ q(n) \end{pmatrix}$$

This defines matrix-vector multiplication.

Photosynthesis



Carbon: $x_1 = 6x_4$

Oxygen: $2x_1 + x_2 = 2x_3 + 6x_4$

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$$x_1 + 0x_2 + 0x_3 - 6x_4 = 0$$

$$2x_1 + x_2 - 2x_3 - 6x_4 = 0$$

$$0x_1 + 2x_2 + 0x_3 - 12x_4 = 0$$

Example 2.2 (Poll everywhere).

$$\begin{pmatrix} 5 & -1 & 0 \\ -3 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix} = ?$$

General matrices and vectors

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m,n}$$

row vector (row matrix): $a = (a_1 a_2 \dots a_n)$

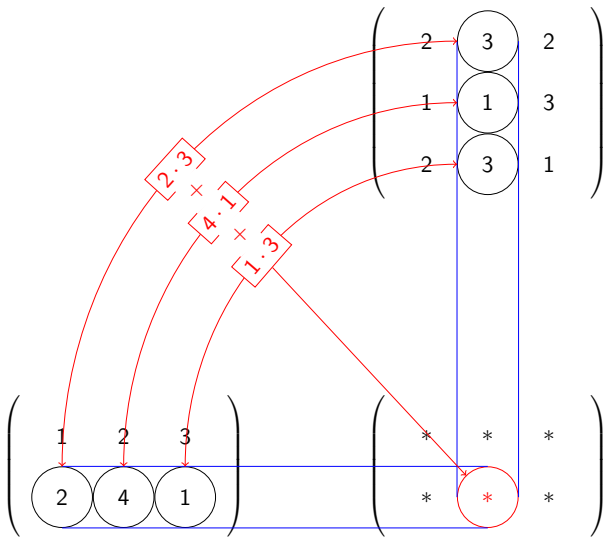
column vector (column matrix): $a = (a_1 a_2 \dots a_n)^\top = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$

Back to herpes simplex example: We have seen that the epidemiological structure of the population has been modelled by

$$\underbrace{\begin{pmatrix} s(n+1) \\ i(n+1) \\ q(n+1) \end{pmatrix}}_{=v(n+1)} = \underbrace{\begin{pmatrix} 0.999946 & 0 & 0 \\ 0.000054 & 0.714 & 0.029 \\ 0 & 0.286 & 0.971 \end{pmatrix}}_{=:T} \underbrace{\begin{pmatrix} s(n) \\ i(n) \\ q(n) \end{pmatrix}}_{=v(n)}$$

Question: Assuming that $v(0)$ is given, what is $v(2)$? Or $v(5)$?
 $\implies$ How to define / calculate $T \cdot T$ respectively T^5 ?

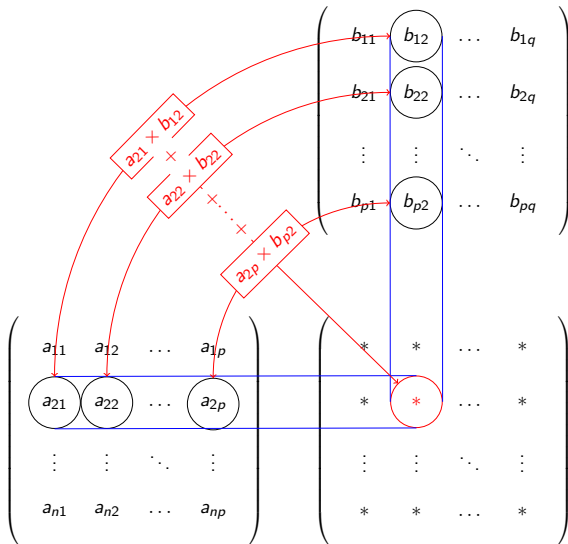
B : 3 rows 3 columns



A : 2 rows 3 columns

$C = A \times B$: 2 rows 3 columns

$B : p \text{ rows } q \text{ columns}$



$A : n \text{ rows } p \text{ columns}$

$C = A \times B : n \text{ rows } q \text{ columns}$

Matrix multiplication

If $A = (a_{ij}) \in \mathbb{R}^{m,n}$ and $B = (b_{jk}) \in \mathbb{R}^{n,o}$, then

$$D = AB \in \mathbb{R}^{m,o}, \quad D = (d_{ik}) \text{ with } d_{ik} = \sum_{j=1}^n a_{ij}b_{jk}.$$

If $C \in \mathbb{R}^{o,p}$, it holds

$$A(BC) = (AB)C = ABC \text{ (associativity of matrix multiplication).}$$

Example 2.3 (Poll everywhere).

Let $A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$. Calculate AB .

Addition of matrices: If $A = (a_{ij}) \in \mathbb{R}^{m,n}$, $B = (b_{ij}) \in \mathbb{R}^{m,n}$, then

$$A + B = (a_{ij} + b_{ij}).$$

It holds for $A \in \mathbb{R}^{m,n}$, $B, C \in \mathbb{R}^{n,o}$

$A(B+C) = AB+AC$ matrix multiplication distributes over addition

Multiplication with a scalar: $cA = (ca_{ij})$

For

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix},$$

find x, λ such that

$$Ax = \lambda x.$$